Parameter Mapping in Doped Inert Cryogenic Crystal Growth: Towards High-Precision Quantum Measurements

Isaac Cross

Abstract:

Doped inert cryogenic crystals offer a promising platform for fundamental physics research, including high-precision quantum measurements and searches for physics beyond the Standard Model. By encasing atoms or molecules (dopants) inside an inert cryogenic crystal, one creates an "artificial vacuum" that enables precise measurements of quantum properties such as the electron electric dipole moment (eEDM) [1]. These crystals also have potential applications in dark matter detection and quantum information science, but current implementations are limited by what many suspect are structural defects [2]. While much research has gone into exploring the use of different dopants and host crystals, and the resulting measurements that can be made, little has been done to fully understand the relationship between crystal growth parameters and resulting crystal structure. Therefore, the goal of this project is to explore this relationship, characterizing how growth rate, dopant density, and beam velocity (beam of dopant and inert gas sent to growth substrate) influence crystal structure and optical clarity. A custom-built dual-species beam source will be used to tune these parameters, growing doped crystals under multiple (~27) distinct conditions spanning the parameter space. Crystal structure and optical clarity will be assessed via optical and x-ray techniques in collaboration with a PhD student, and a detailed, systematic mapping will be made linking growth conditions to crystal properties in these systems. This study will develop insight into the formation and structure of doped cryogenic crystals, which is critical for achieving their full potential in fundamental physics.

Where Work Will Be Done and New Knowledge Produced:

This research will be conducted in the Odom Lab, located within the Center for Fundamental Physics at Northwestern University. Within this lab, I have already designed and am assembling the dual-species beam source that co-delivers rubidium and argon atoms to the growth substrate. This beam source has control over the growth parameters I wish to study in this project, so characterization of this beam is critical for understanding growth parameters. All the tools necessary for this task and initial crystal characterization are already available in our lab on Northwestern's Campus.

The new knowledge produced by this research will be a systematic mapping of how growth parameters–beam velocity, dopant density, and growth rate–affect the structure and optical clarity of rubidium-doped argon crystals. Although past research has focused on different dopants or host materials, little has been done to systematically study the relationship between growth parameters and the resulting crystal structure. The work stands to uncover which growth regimes produce homogeneous crystals with high optical clarity—key for enabling the use of these crystals in high precision quantum measurements.

Work Description:

The motivation to study the growth of these crystals stems from their strong potential in high precision measurement experiments. Doped cryogenic crystals enable a vast increase in

statistical sensitivity, as they can isolate a large number of atoms of molecules for simultaneous measurements. For instance, one of the most competitive eEDM measurements, by ACME-II, required 350 hours of data collection with a molecular beam of a few hundred thousand molecules per second [3]. In contrast, a cryogenic crystal could host 10^{13} dopants/mm³ and produce an eEDM measurement with the same uncertainty in just seconds rather than weeks [7]. To do this, however, the dopants need to maintain a high enough 'ensemble transverse spin relaxation time' otherwise known as T2* or a coherence time (the time before a quantum system loses stability from environmental effects). With the dopant density previously mentioned, T2* must be at least a ms. While previous work has achieved T2* ~ 1 ms with cesium-doped helium crystals, the extreme pressures required to solidify helium makes high dopant density technically intractable [4]. Other systems–such as neon, parahydrogen, and argon crystals doped with alkali-atoms–have achieved the desired dopant densities but with insufficient T2* times (~10-100us) [2, 5, 6]. Many claim the dominant limitation on coherence time comes from crystal inhomogeneities as dopants which experience different local geometry inside the same crystal will dephase, decreasing the size of the signal [2]. Embedding dopants in a perfect single crystal could mitigate this issue, but a reliable method for growing such crystals at sufficient dopant densities has not yet been demonstrated. It is the goal of the Odom Lab to address these issues and be the first to grow a perfect single crystal with sufficient doping and T2* times.

As part of this larger project, my project seeks to investigate how key growth parameters influence structural and optical quality of rubidium doped argon crystals–optical clarity is essential for implementing optical measurement techniques which are integral to precision measurement experiments like eEDM searches. The focus is on three controllable variables: beam velocity, dopant density, and growth rate. Beam velocity affects how dopants incorporate into the lattice during growth; excessive velocity may lead to disordered embedding, while too slow a beam may result in clustering. Dopant density determines interatomic spacing, which is critical for coherence, but also increases the risk of dipolar interactions and decoherence if the crystal is not uniform. Growth rate influences how atoms arrange themselves on the substrate—many groups adopt values around 0.5–2 μm/min without much explanation [10, 8]. Together, these parameters define a space in which optimized growth conditions may exist, but this space remains unexplored. This project will create the first detailed parameter map connecting multiple beam conditions to measurable crystal outcomes, with the eventual goal of identifying regimes that support homogeneous, optically clear crystals capable of long T2*.

Therefore, this project will proceed in two phases: (1) characterization of a dual-species beam source and (2) growth and analysis of rubidium-doped argon crystals under systematically varied growth conditions.

In phase one, beam velocity will be determined by observing what's known as Doppler shifts in absorption spectroscopy: as rubidium atoms move toward or away from a probe laser, their transition frequencies shift proportionally to their velocity [9]. Argon flow rate, measured via calibrated flow meters and system geometry, will determine overall growth rate. Dopant density can be determined using absorption spectroscopy, which, with the beam velocity measurements, tells us the flow rate of the rubidium atoms. The ratio between this and argon flow rate is the dopant density [10]. The dual-species beam source I designed allows tunability over a wide range of conditions: growth rates from 0.1 to 10 μm/min, dopant densities from 10^{11} to 10^{15} dopants/mm³, and velocities from 250 to 600 m/s. Initially, I will study three values for each

parameter–spanning low, medium, and high values relative to and including parameters used by other groups–resulting in 27 unique growth conditions. The goal of phase one is to identify how beam source settings map to physical growth parameters.

In phase two, I will use the identified parameters to grow crystals, while a PhD student in our lab will characterize their structure and optical clarity using optical and x-ray techniques. Although I will largely assist in this analysis, my focus remains on mapping the growth parameter space to the resulting crystal properties measured.

Value of Award:

Receiving the McCormick Summer Research Award would allow me to devote a full summer to pursuing a project that sits at the intersection of my academic interests and long-term career goals. I aim to pursue a PhD in physics focusing on precision quantum measurements and fundamental physics. This project will be a valuable opportunity to deepen my experience in a field I hope to continue in and push me to develop techniques and knowledge not always accessible to undergraduates. The award would give me the time and independence to bring a complex apparatus into operation, explore a broad parameter space, and generate new data that will contribute to both my lab's broader research effort and my own senior thesis. Additionally, working closely with a PhD student on the crystal analysis will give me valuable experience in scientific collaboration and help prepare me for the collaborative environment of experimental physics.

References:

Note: Citations in APS style

[1] A. C. Vutha, M. Horbatsch, and E. A. Hessels. Oriented polar molecules in a solid inert-gas matrix: a proposed method for measuring the electric dipole moment of the electron. *Atoms*, 6(1), 3 (2018).

[2] S. Upadhyay, U. Dargyte, V. D. Dergachev, R. P. Prater, S. A. Varganov, T. V. Tscherbul, D. Patterson, and J. D. Weinstein. Spin coherence and optical properties of alkali-metal atoms in solid parahydrogen. *Physical Review A*, 100(6), 063419 (2019).

[3] ACME. Improved limit on the electric dipole moment of the electron. *Nature*, 562(7727), 355–360 (2018).

[4] S. I. Kanorsky, S. Lang, S. Lucke, S. B. Ross, T. W. Hänsch, and A. Weis. Millihertz magnetic resonance spectroscopy of Cs atoms in body-centered-cubic ^{4}He. *Physical Review A*, 54, R1010–R1013 (1996).

[5] D. M. Lancaster, U. Dargyte, S. Upadhyay, and J. D. Weinstein. Radiative properties of rubidium atoms trapped in solid neon and parahydrogen. *Physical Review A*, 106, 042810 (2022).

[6] U. Dargyte, D. M. Lancaster, and J. D. Weinstein. Optical and spin-coherence properties of rubidium atoms trapped in solid neon. *Physical Review A*, 104(3), 032611 (2021).

[7] B. Odom. PM: Doped inert single crystals for probing beyond the standard model. *National Science Foundation*, (2022).

[8] U. Dargyte, D. M. Lancaster, and J. D. Weinstein. *Optical and Spin-Coherence Properties of Rubidium Atoms Trapped in Solid Neon*. Manuscript in preparation (2021).

[9] D. O. Sabulsky. *A Cesium-133 Effusive Oven for Ultracold Atomic Experiments*. Bachelor's thesis, The University of Chicago (2014).

[10] S. K. Regmi. *Optical Pumping of Rubidium Spin in a Solid Argon Matri*x. PhD thesis, University of Nevada, Reno (2013).

Personal Resume:

I have worked in the Odom Lab at Northwestern since Fall 2023, where I designed and began assembling a dual-species atomic beam source for growing rubidium-doped argon crystals. I have also contributed to the design of the main experimental chamber used for cryogenic crystal growth and characterization. Through this work, I've gained experience with ultra-high vacuum systems, optical components, and laser safety, and have written standard operating procedures (SOPs) for our group. I work closely with a PhD student in the lab and present updates in our weekly research meetings with Professor Odom. Prior to this, I conducted research in the Kovachy Lab, where I gained experience in machining, CAD modeling, and hardware design. I collaborated with graduate students and built components for cold atom interferometry experiments, further strengthening my technical foundation in experimental physics.

By this summer, I will have completed all of my Physics major requirements, including two advanced laboratory courses that developed my skills in experimental design, data analysis, and critical thinking. These experiences have prepared me to work independently and to understand the physical principles underlying the techniques used in this project. Beyond my own lab, I also communicate frequently with students in the Kovachy Lab and others in our research center, building a strong peer network in experimental AMO (atomic, molecular, and optical) physics I can rely on.